

目的: Configure SR-IOV in RHEL

RHEL 9 & Intel I350-T4

HOST configuration:

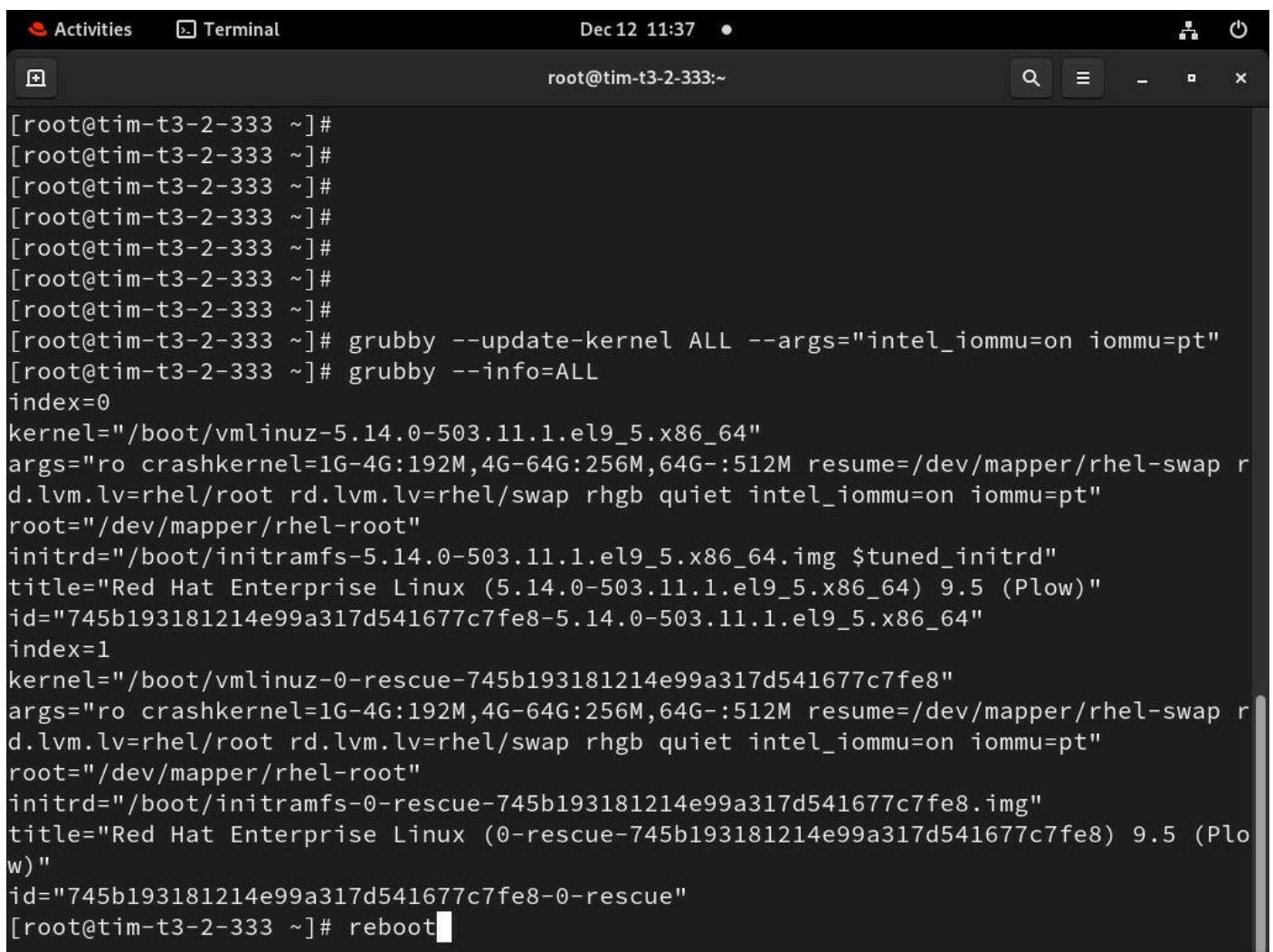
step 1. Add kernel parameter

But to get you started, once you have enabled `iommu=pt` and `intel_iommu=on` as kernel boot parameters, if you are running a Linux kernel that is at least 3.8.x, to initialize SR-IOV VFs issue the following command:

```
#grubby --update-kernel ALL --args="intel_iommu=on iommu=pt"
```

```
#grubby --info=ALL
```

```
#reboot
```

A screenshot of a terminal window titled "Terminal" with a timestamp of "Dec 12 11:37". The terminal shows the execution of the command `grubby --info=ALL` to display kernel boot parameters. The output lists two kernel entries: the main kernel at index 0 and a rescue kernel at index 1. Both entries show the same arguments, including `intel_iommu=on` and `iommu=pt`. The terminal prompt is `root@tim-t3-2-333:~`.

```
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]#  
[root@tim-t3-2-333 ~]# grubby --update-kernel ALL --args="intel_iommu=on iommu=pt"  
[root@tim-t3-2-333 ~]# grubby --info=ALL  
index=0  
kernel="/boot/vmlinuz-5.14.0-503.11.1.el9_5.x86_64"  
args="ro crashkernel=1G-4G:192M,4G-64G:256M,64G-:512M resume=/dev/mapper/rhel-swap r  
d.lvm.lv=rhel/root rd.lvm.lv=rhel/swap rhgb quiet intel_iommu=on iommu=pt"  
root="/dev/mapper/rhel-root"  
initrd="/boot/initramfs-5.14.0-503.11.1.el9_5.x86_64.img $tuned_initrd"  
title="Red Hat Enterprise Linux (5.14.0-503.11.1.el9_5.x86_64) 9.5 (Plow)"  
id="745b193181214e99a317d541677c7fe8-5.14.0-503.11.1.el9_5.x86_64"  
index=1  
kernel="/boot/vmlinuz-0-rescue-745b193181214e99a317d541677c7fe8"  
args="ro crashkernel=1G-4G:192M,4G-64G:256M,64G-:512M resume=/dev/mapper/rhel-swap r  
d.lvm.lv=rhel/root rd.lvm.lv=rhel/swap rhgb quiet intel_iommu=on iommu=pt"  
root="/dev/mapper/rhel-root"  
initrd="/boot/initramfs-0-rescue-745b193181214e99a317d541677c7fe8.img"  
title="Red Hat Enterprise Linux (0-rescue-745b193181214e99a317d541677c7fe8) 9.5 (Plo  
w)"  
id="745b193181214e99a317d541677c7fe8-0-rescue"  
[root@tim-t3-2-333 ~]# reboot
```

step 2. 查看 device name，以確認要測試的 port 代號

```
#ifconfig
```

```
enp225s0f0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

```
inet 192.168.1.2 netmask 255.255.255.0 broadcast 192.168.1.255
```

```
inet6 fe80::8da4:bdb:7684:5285 prefixlen 64 scopeid 0x20<link>
```

```
ether b4:96:91:e8:91:80 txqueuelen 1000 (Ethernet)
```

RX packets 24592116 bytes 37232205666 (34.6 GiB)

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 2027579 bytes 133828240 (127.6 MiB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

device memory 0xba300000-ba3fffff

```
Activities Terminal Dec 12 11:45
root@tim-t3-2-333:~
inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
    valid_lft forever preferred_lft forever
[root@tim-t3-2-333 ~]#
[root@tim-t3-2-333 ~]# ifconfig
enol: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.101.43 netmask 255.255.255.0 broadcast 192.168.101.255
    inet6 fe80::c67d:46ff:fec4:4e33 prefixlen 64 scopeid 0x20<link>
    ether c4:7d:46:c4:4e:33 txqueuelen 1000 (Ethernet)
    RX packets 255366 bytes 29157317 (27.8 MiB)
    RX errors 0 dropped 100587 overruns 0 frame 0
    TX packets 8984 bytes 692308 (676.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device memory 0xaca00000-aca7ffff

enp225s0f0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.2 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::8da4:bdb:7684:5285 prefixlen 64 scopeid 0x20<link>
    ether b4:96:91:e8:91:80 txqueuelen 1000 (Ethernet)
    RX packets 24592116 bytes 37232205666 (34.6 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2027579 bytes 133828240 (127.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device memory 0xba300000-ba3fffff

enp225s0f1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether b4:96:91:e8:91:81 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device memory 0xba200000-ba2fffff

enp225s0f2: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
```

step 3. Set up SR-IOV LAN VF funtion

```
#echo 2 > /sys/class/net/<device name>/device/sriov_numvfs
```

請注意這個方式是在一張卡的一個實體網路 port, 開啟兩個 VF function

step 3. Check VF adapters

Once an SR-IOV NIC VF is created on the host, the driver/OS assigns a MAC address and creates a network interface for the VF adapter.

```
#ifconfig
```

```
enp225s0f0v0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
```

```
    ether 72:b8:e2:1d:34:11 txqueuelen 1000 (Ethernet)
```

```
    RX packets 0 bytes 0 (0.0 B)
```

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 0 bytes 0 (0.0 B)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp225s0f0v1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500

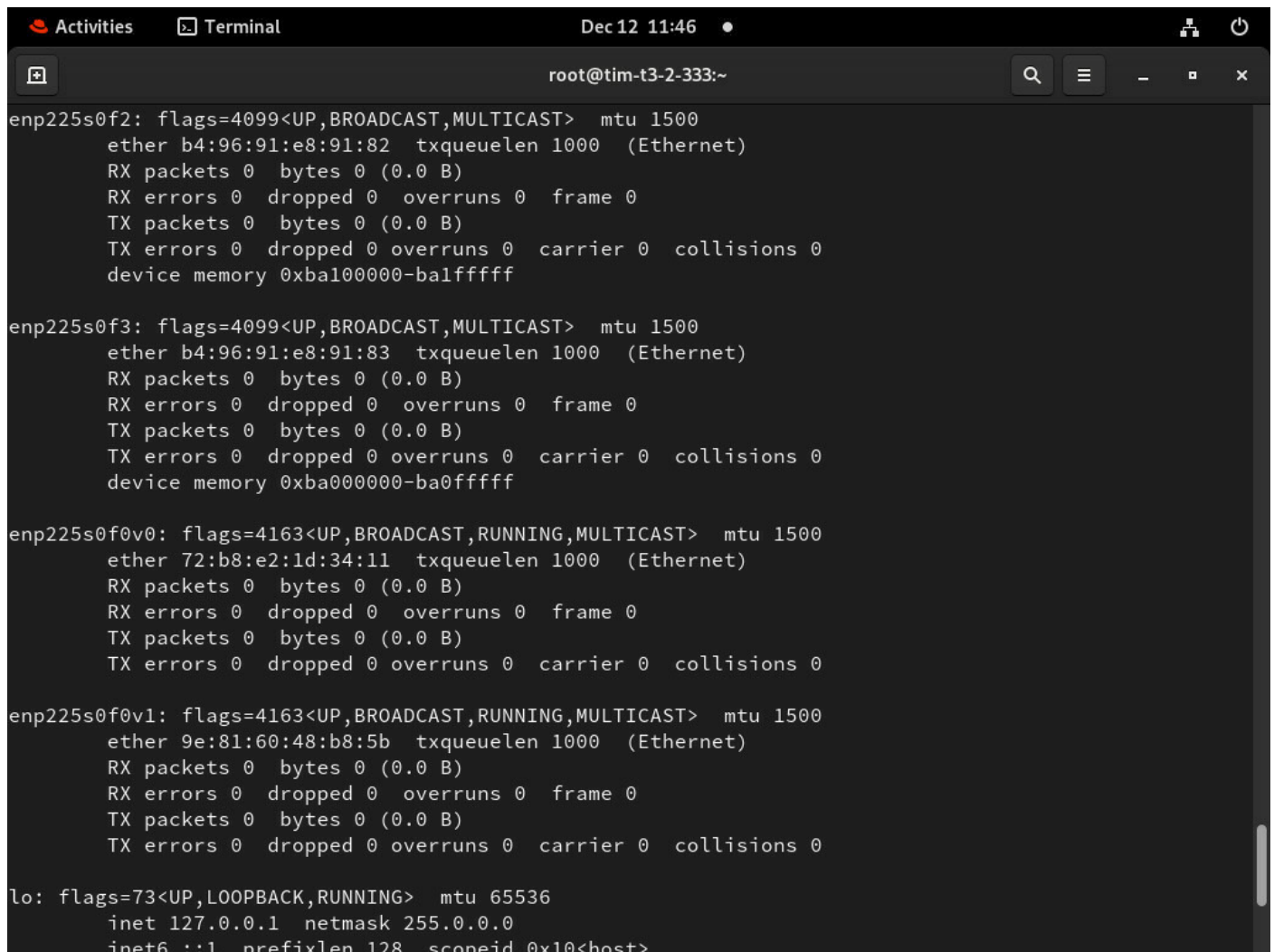
ether 9e:81:60:48:b8:5b txqueuelen 1000 (Ethernet)

RX packets 0 bytes 0 (0.0 B)

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 0 bytes 0 (0.0 B)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0



```
Activities Terminal Dec 12 11:46 root@tim-t3-2-333:~
enp225s0f2: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether b4:96:91:e8:91:82 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device memory 0xba100000-ba1fffff

enp225s0f3: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether b4:96:91:e8:91:83 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device memory 0xba000000-ba0fffff

enp225s0f0v0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 72:b8:e2:1d:34:11 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

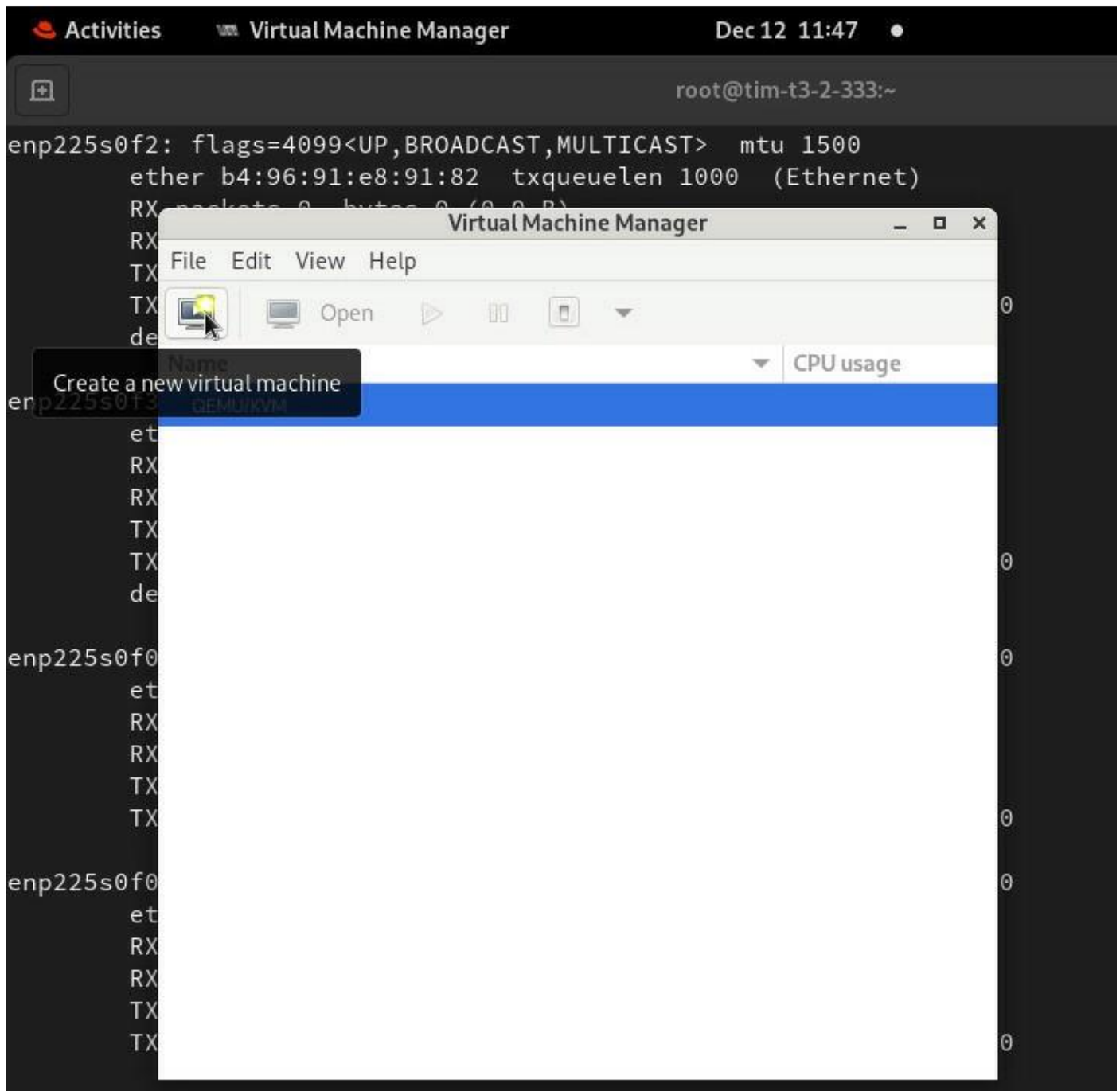
enp225s0f0v1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 9e:81:60:48:b8:5b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

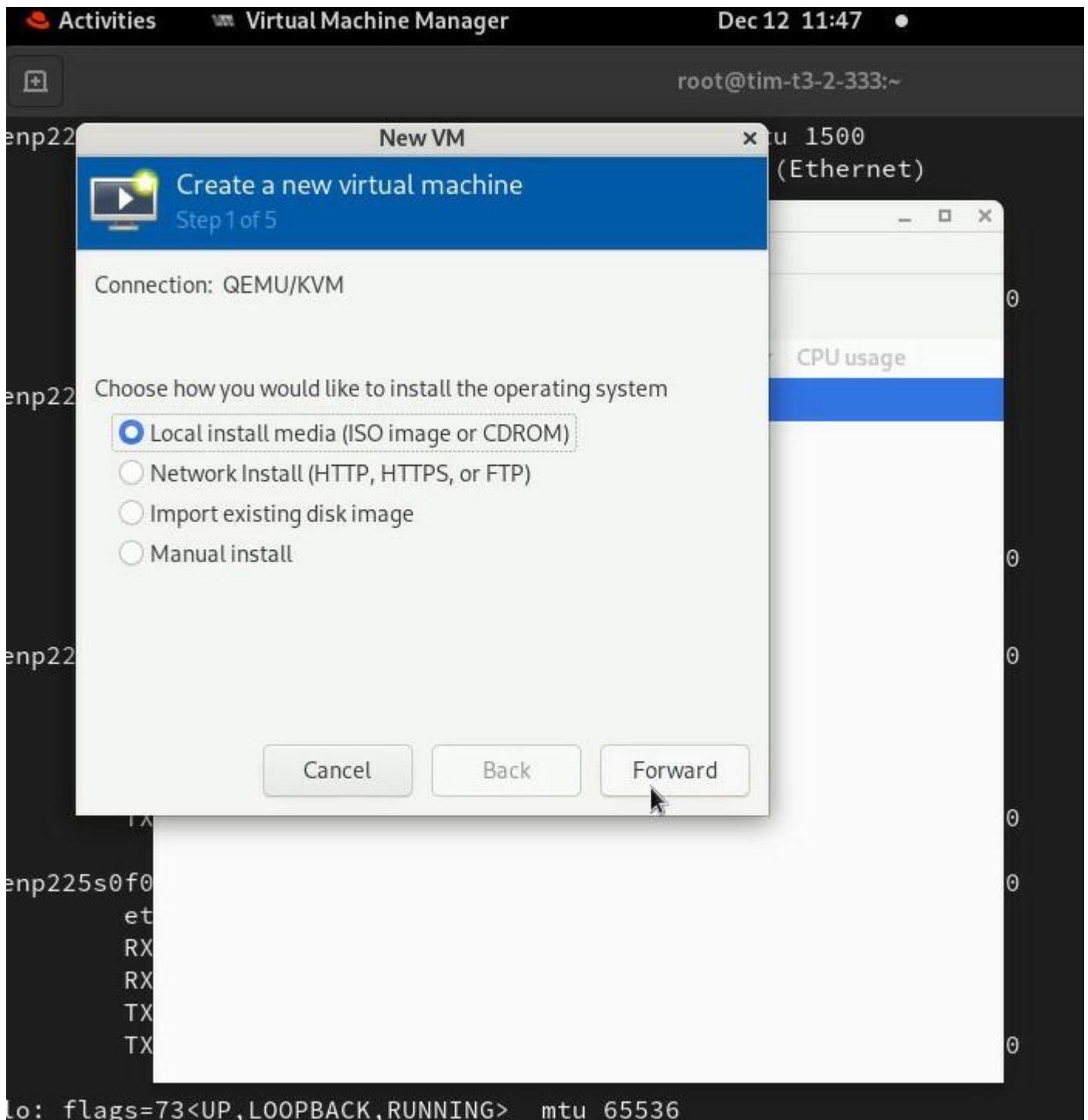
step 4~14 Add hardware device by Virtual Machine Manager.

SR-IOV Virtual Function PCI Passthrough Device

step 4. Open the "Virtual Machine Manager" and click on "Create a New Virtual Machine."



step 5. Select "Local install media (ISO)" and click "Forward" to proceed to the next step.



step 6~7. Click "Browse," then click "Browse Local" to select the pre-prepared OS ISO file.

New VM

Create a new virtual machine
Step 2 of 5

Choose ISO or CDROM install media:

No media selected

Browse...

Choose the operating system you are installing:

Waiting for install media / source

☒ Automatically detect from the installation media / source

Cancel Back Forward

enp225s0f0

et

RX

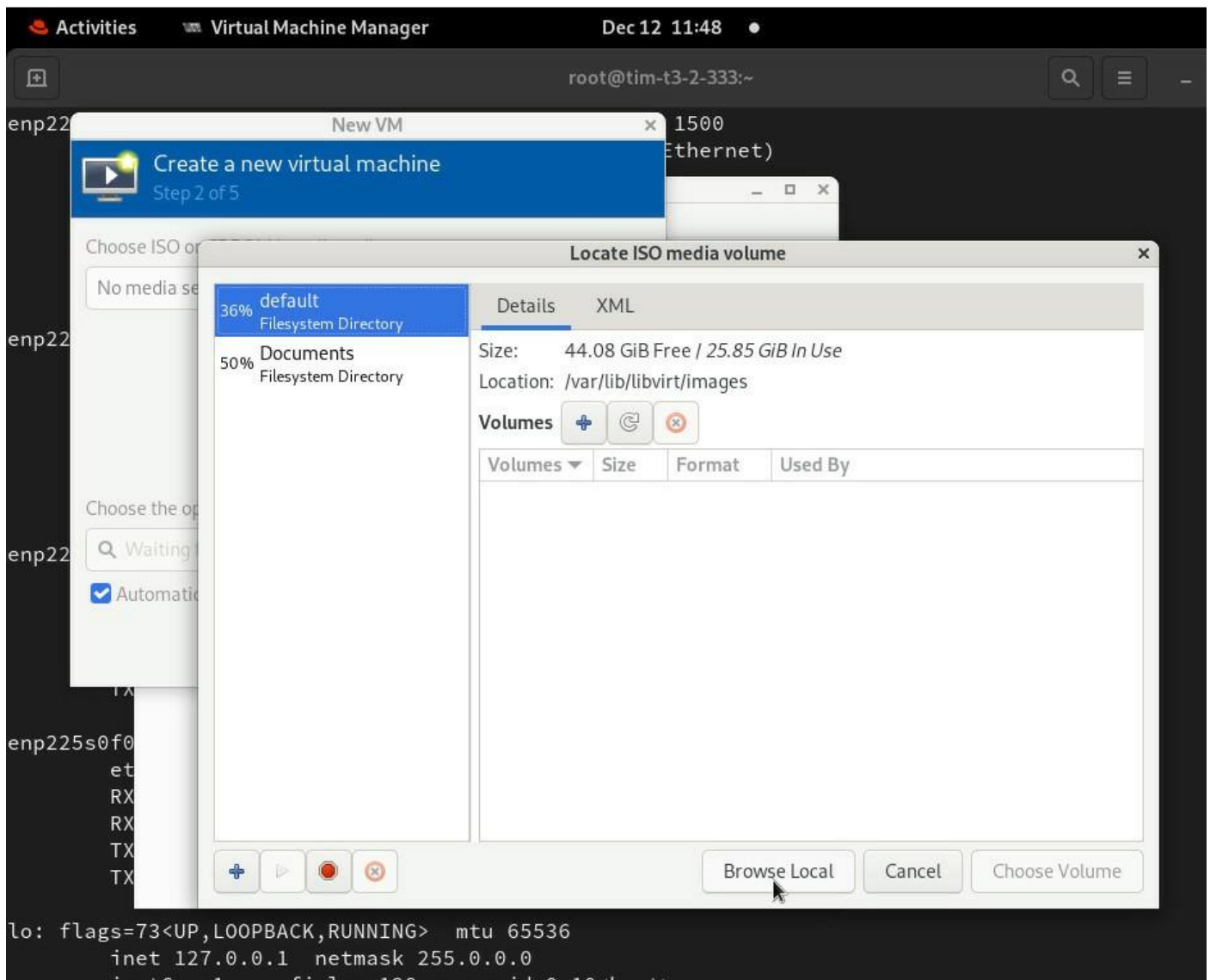
RX

TX

TX

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536

inet 127.0.0.1 netmask 255.0.0.0

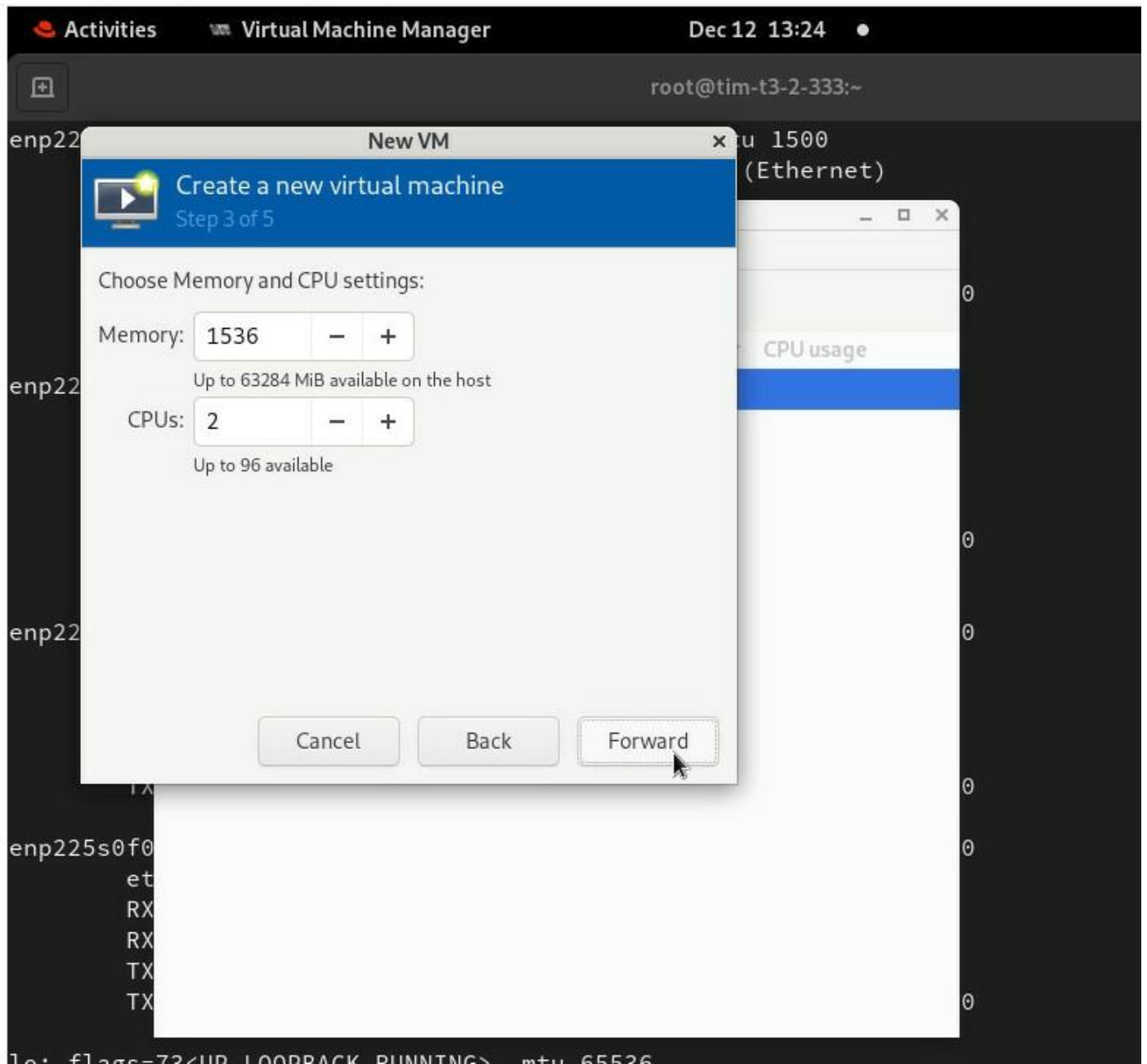


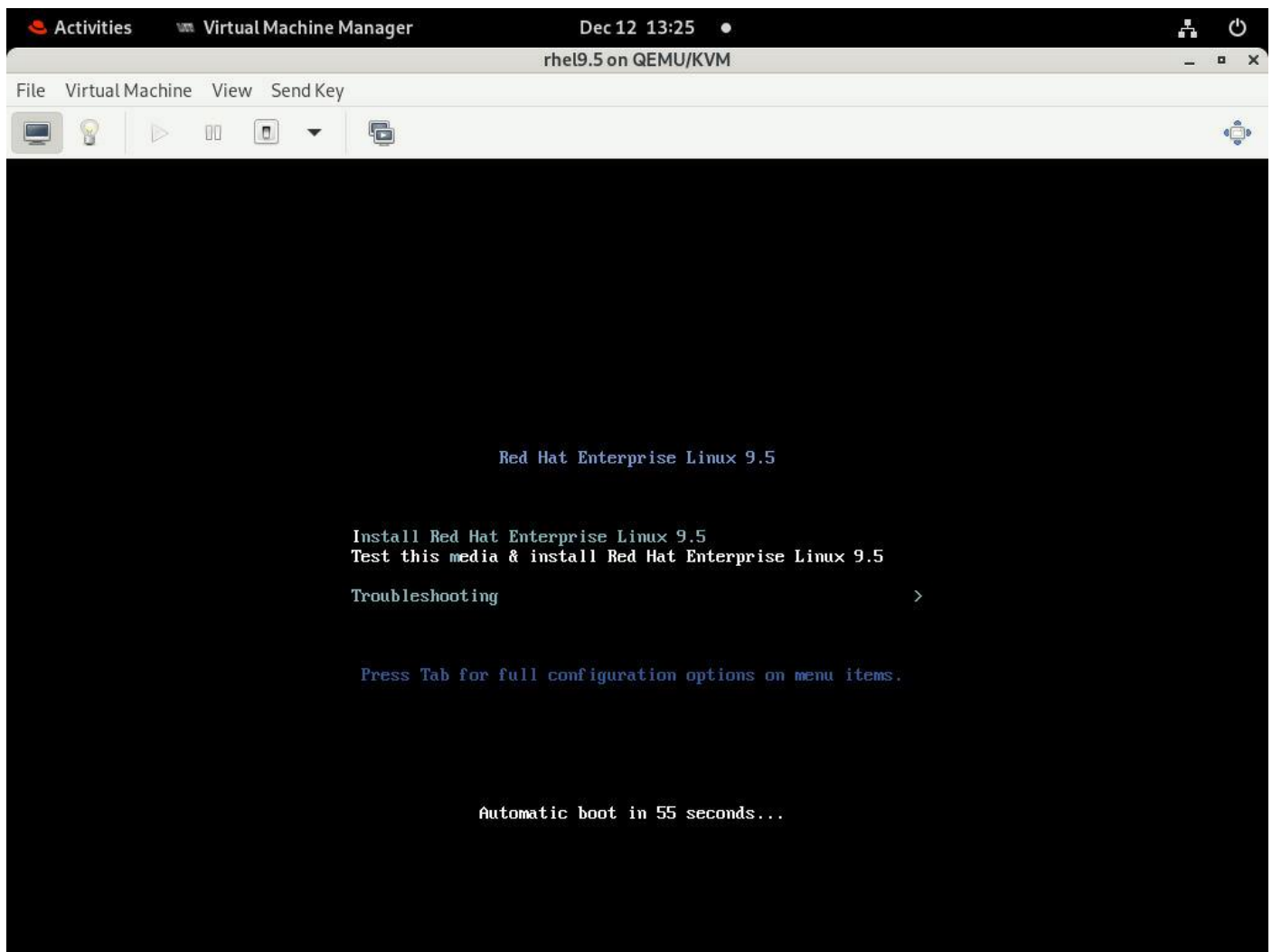
step 8~11. Adjust the configuration of the virtual machine and proceed with the OS installation.



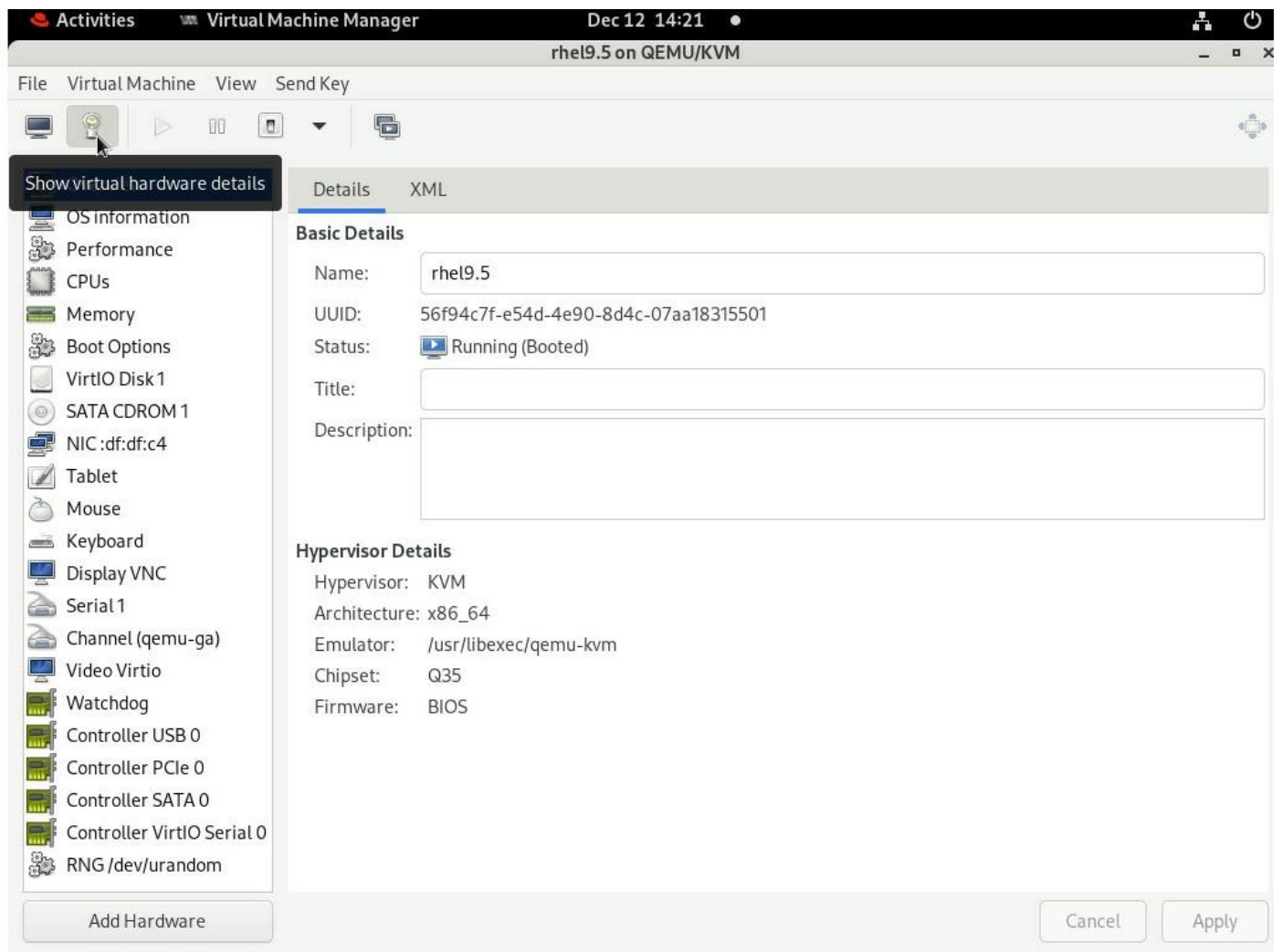
CPU usage

enp225s0f0
et
RX
RX
TX
TX

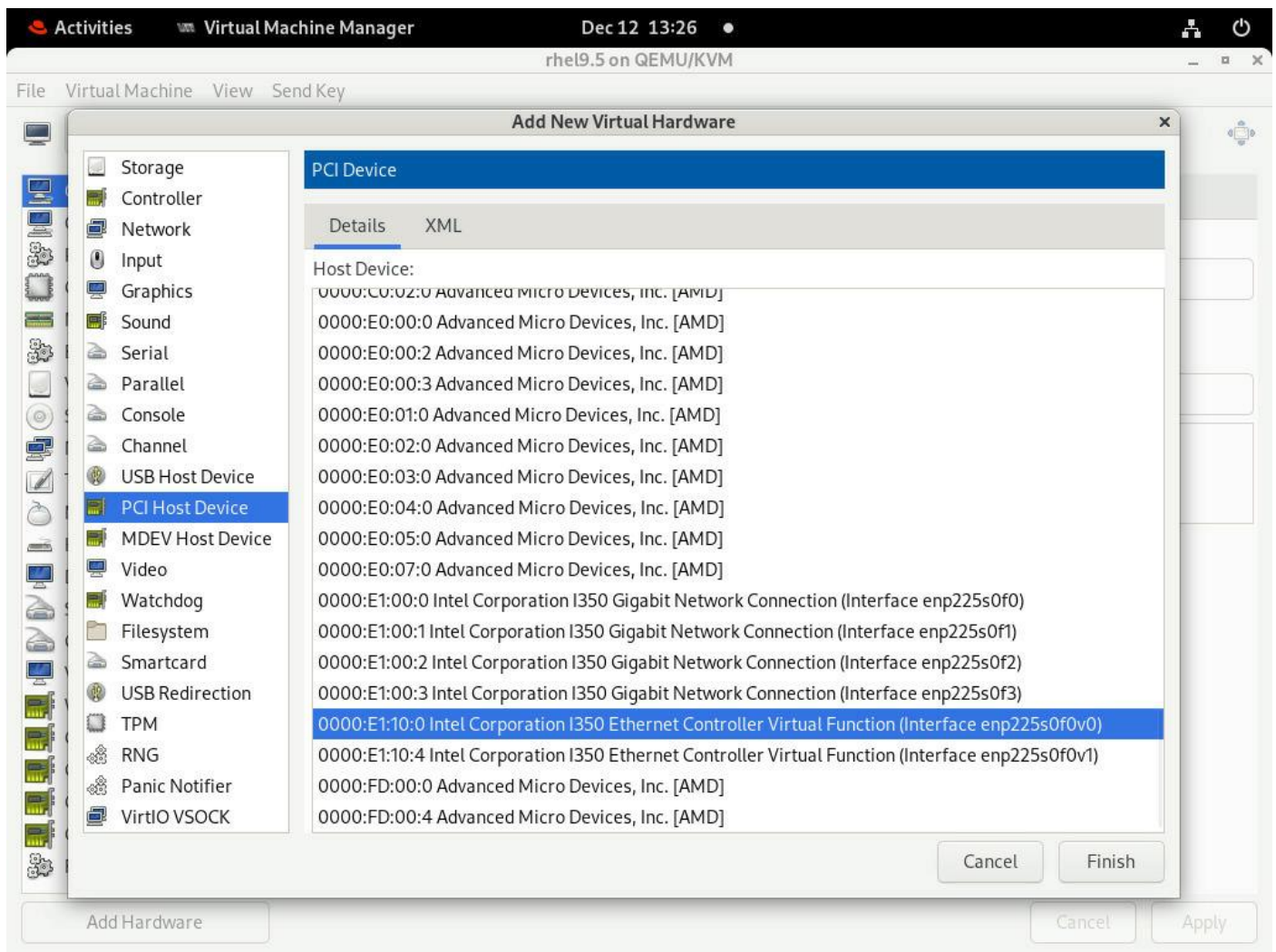




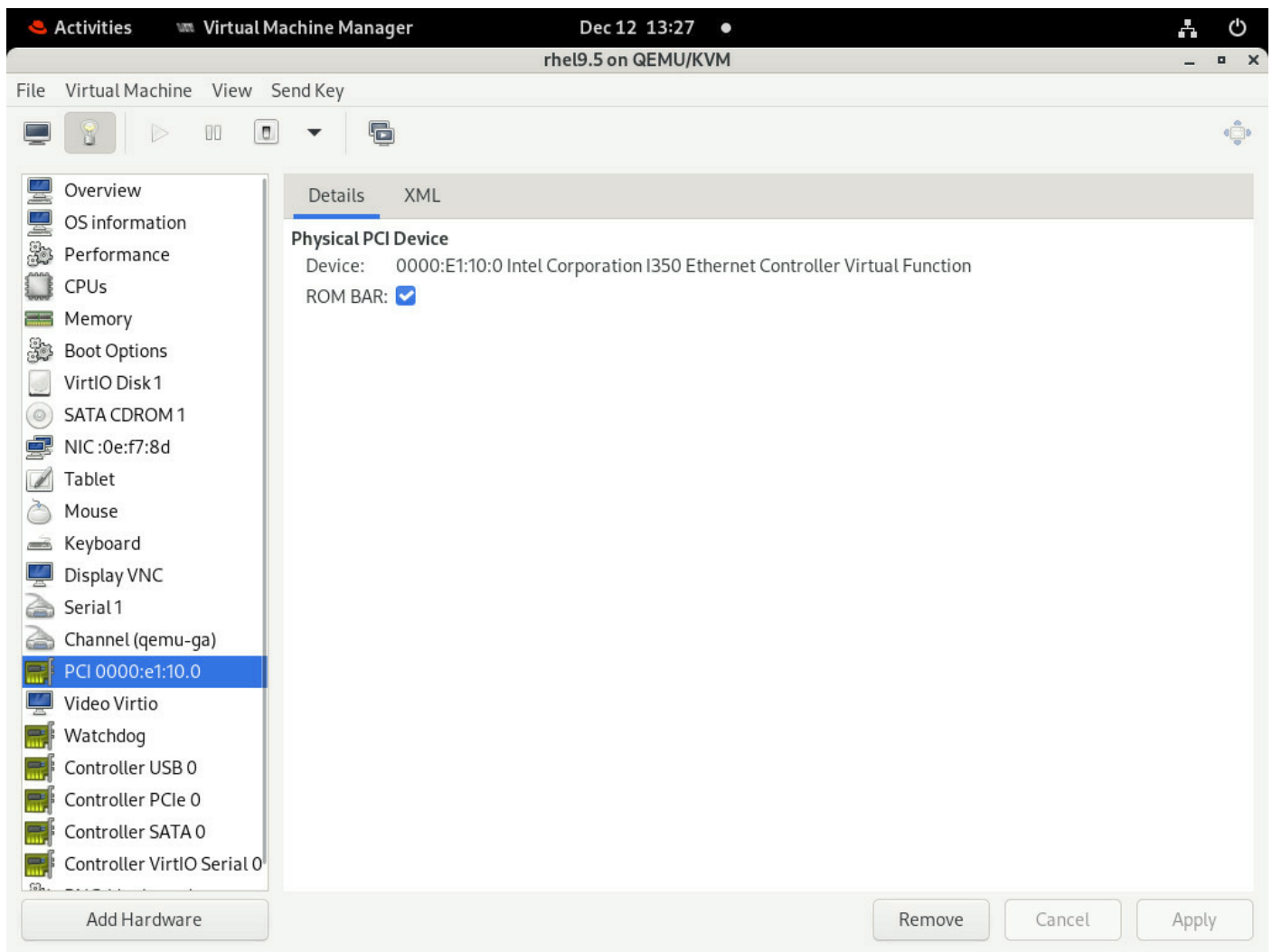
step 12. Click "Show virtual hardware details" and then click "Add Hardware" to select the hardware you want to add to the virtual machine.



step 13. Click "PCI Host Device" and then select the virtual network port created in Step 3.



step 14. Verify if the addition was successful.



step 15. Enter the Virtual Machine's OS and verify the throughput.